INCH-POUND

MIL-I-43684B
30 June 1989
SUPERSEDING
MIL-I-43684A
29 December 1980

MILITARY SPECIFICATION

INTRENCHING TOOL, HAND, FOLDING, LIGHTWEIGHT

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type and size of lightweight, foldable, hand intrenching tool (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5120

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SPECIFICATIONS

FEDERAL

PPP-B-601	-	Boxes, Wood, Cleated-Plywood
PPP-B-640	-	Box, Fiberboard, Corrugated, Triple-Wall
PPP-C-843	-	Cushioning Material, Cellulosic
PPP-P-291	-	Paperboard, Wrapping and Cushioning

MILITARY

MIL-H-6088	-	Heat Treatment of Aluminum Alloys

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-130	-	Identification Marking of U.S. Military Property
MIL-STD-147	-	Palletized Unit Loads
MIL-STD-731	-	Quality of Wood Members for Containers and Pallets

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

US ARMY NATICK RESEARCH AND DEVELOPMENT LABORATORIES

2-2-284	-	Intrenching Tool, Hand, Folding, Lightweight; Assembly Complete
2-2-285	-	Intrenching Tool, Hand, Folding, Lightweight; Details A
2-2-286	-	Intrenching Tool, Hand, Folding, Lightweight; Details B
2-2-288	-	Intrenching Tool, Hand, Folding, Lightweight; Strength Test

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(Copies of drawings are available from the U.S. Army Natick Research, Development, and Engineering Center, ATTN: STRNC-EMSS, Natick, MA 01760-5014.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

E 18 - Metallic Materials,Superficical Hardness, Tests for Rockwell

D 3951 - Standard Practice for Commercial Packaging

(Applications for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3), in accordance with 4.3.

3.2 Materials and components. Materials and components shall be as specified on the applicable drawings, and as specified herein. For materials or components for which it is stated "or equal", if the contractor proposes to use an item considered to be equal to the materials or component specified, prior to its use the contractor shall furnish a sample of the material or component, with data to the contracting officer for subsequent evaluation by the responsible military agency. The supporting data required shall prove the functional equivalence and design compatibility of the item proposed to be used. It is encouraged that recycled material be used when practical as long as it meets the requirements of this specification.

3.3 Design and construction. The intrenching tool shall conform to the design, details, dimensions and materials specified on Drawing No. 2-2-284, 2-2-285, 2-2-286, and 2-2-288, and as specified herein. The blade portion of a finished intrenching tool shall move on the hinges to picking and shoveling

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positions when tested as specified in 4.5.1. The intrenching tool shall show no fracture of components, and shall have no permanent deflection in excess of 5/16 inch when tested as specified in 4.5.2. The hinge and the rear handle shall bear evenly on the thrust washer and thrust plate, respectively, in operating position.

3.3.1. Riveting.

3.3.1.1 Hinge to blade. Hinge shall be riveted to the blade. When the solid rivets are driven, they shall completely fill the holes. The rivet heads shall be full, neatly made, concentric with the rivet hole, and shall be in full contact with the surface of the riveted member.

3.3.1.2 Semi-tubular rivet. The end of the rivet shall be orbitally spun as shown on drawing 2-2-284 and shall be concentric with the rivet hole.

3.3.2 Heat treatment.

3.3.2.1 Steel components. Steel components, except for rivets and spring, shall be heat treated as shown on applicable drawings. Hardness shall be determined in accordance with 4.5.3.

3.3.2.2 Aluminum components. Aluminum components shall be heat treated to the tempers specified on the applicable drawings and in accordance with MIL-H-6088.

3.4 Finish.

3.4.1 Steel components. Prior to assembly, steel components shall be either phosphate coated (see 6.4) or pretreated for painting in accordance with the requirements specified on the applicable drawings.

3.4.1.1 Enameling. After pretreatment, those components for which painting is specified, shall be coated with enamel as specified on the applicable drawing. The thrust washer and nut shall be enameled after bonding.

3.4.2 Aluminum components. Prior to assembly, the aluminum components, except the spring pin, shall be annodized as specified on the applicable drawing.

3.5 Bonding. The thrust washer and nut shall be thoroughly cleaned and bonded to withstand the test in 4.5.4.

3.6 Lubrication. Prior to assembly of the nut to the handle the threads of each shall be lightly coated with the lubricant specified on drawing 2-2-284. The hinge shall be lubricated as specified in drawing 2-2-284.

3.7 Marking for identification. The intrenching tool shall be marked for identification as shown on drawing 2-2-285 and shall be in accordance with MIL-STD-130.

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3.8 Workmanship. The surface of the intrenching tool shall be free of burrs, slivers, sharp corners, and rough or sharp edges, except for the serrated cutting edges of the blade. The feather edge shall be removed from the beveled, serrated cutting edges. The coating shall be a continuous, uniform, smooth, dry film without any area of no film, and shall be free from embedded foreign matter. The intrenching tool shall conform to the quality of product established by this specification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Responsibility for dimensional requirements. Unless otherwise specified in the contract or purchase order, the contractor is responsible for ensuring that all specified dimensions have been met. When dimensions cannot be examined on the end item, inspection shall be made at any point, or at all points in the manufacturing process necessary to assure compliance with all dimensional requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3)
b. Quality conformance inspection (see 4.4)

4.3 First article inspection. When a first article inspection is required (see 3.1 and 6.2), it shall be examined for the defects specified in 4.4.3 and 4.4.4 and tested as specified in 4.4.5.

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4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.4.1.1 Component testing. Prior to assembly, the blade, hinges and thrust washer shall be tested in accordance with 4.5.3 to determine compliance with 3.3.2.1. The hinges, thrust washers, and blades shall be tested separately. The lot size shall be expressed as all the blades, hinges, or thrusts washers, as applicable, heat treated as a group and offered for inspection at one time. The sample unit shall be one blade, one hinge or one thrust washer. After heat treatment and bonding of thrust washer to nut, the sub-assembly shall be tested as specified in 4.5.4 to determine compliance with 3.5. The inspection level for both tests shall be S-2 (see 6.5).

4.4.2 In-process Inspection. Inspection shall be made, during the manufacturing process, for the requirements specified in table I, to establish that no deviation is made from indicated requirements. Whenever nonconformance is noted, correction shall be made to the affected items and process. The Government reserves the right to exclude from consideration for acceptance, any material or service for which in-process inspection has indicated nonconformance.

TABLE I. In-process inspection

Requirement	Requirement paragraph
Blade movement	3.3
Rivet holes filled	3.3.1.1
Pretreatment of steel components prior to testing	3.4.1
Anodizing of aluminum components prior to enameling	3.4.2
Pre-bonding preparation	3.5

4.4.3 End item visual examination. The end items shall be examined for the defects listed in table II. The lot size shall be all of intrenching tools offered for inspection at on time. The sample unit shall be one intrenching tool. The inspection level shall be II (see 6.5).

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TABLE II. End item visual defects

Examine	Defect	Classification Major	Minor
Finish	Not type specified	101	
	Color not as specified		201
	Area of no paint		202
	Contains streaks, runs, drops, grit or wrinkles		203
	Dirt or other foreign matter imbedded in finish		204
Construction	No lubricant on threads of nut and handle	102	
	Component not type specified	103	
	Hinge does not bear evenly on thrust washer in operating position	104	
	Rear handle does not bear evenly on thrust plate in operation position	105	
	Handle assembly off center	106	
	Serrated and bevel edge missing	107	
	Surface not free from burrs, slivers, sharp corners and rough or sharp edges, except for the beveled, serrated cutting edges	108	
	Feather edge not removed from beveled, serrated cutting edge		205
	Any rivet missing	109	
	Rivet not peened, curled, or end not mushroomed	110	
	Rivet head missing	111	
	Any rivet reversed, e.g., head is not on the intended side of component as required		206
	Rivets not in full contact with surface of riveted member	112	
	Threads on locking nut and outer handle crossed or damaged	113	
Identification marking	Omitted, incorrect, illegible, misplaced, or size of characters not as specified		207

4.4.4 End item dimensional examination. The lot size shall be examined for conformance to the dimensions specified on the drawings. Only those dimensions that can be evaluated without damaging or disassembling the end items shall be examined. Any dimension not within the specified tolerance shall be classified

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as a defect. The lot size shall be expressed in units of intrenching tools. The sample unit shall be one intrenching tool. The inspection level shall be S-3 (see 6.5).

4.4.5 End item testing. The intrenching tool shall be tested in accordance with 4.5.1 and 4.5.2. For the test in 4.5.1, a single sampling plan expressed in terms of defects per hundred units shall be applicable. For the test in 4.5.2 a double sampling plan shall be used. The inspection level shall be S-2 (see 6.5).

4.4.6 Packaging examination. The fully packaged end items shall be examined for the defects listed below. The lot size shall be expressed in units of shipping containers. The sample unit shall be one shipping container fully packaged. The inspection level shall be S-2 (see 6.5).

Examine	Defect
Marking (exterior and interior)	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application.
Materials	Any component missing, damaged, or not as specified.
Workmanship	Inadequate application of components, such as: incomplete sealing or closure of flap, improper taping, loose strapping or inadequate stapling. Bulged or distorted container.
Content	Number per container is more or less than required.

4.4.7 Palletization examination. The fully packaged and palletized end items shall be examined for the defects listed below. The lot size shall be all of the palletized units offered for inspection at one time. The sample unit shall be one palletized unit load, fully packaged. The inspection level shall be S-1 (see 6.5).

Examine	Defect
Finished dimensions	Length, width, or height exceeds specified maximum requirements
Palletization	Pallet pattern not as specified Interlocking of loads not as specified Load not bonded as specified

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Examine	Defects
Weight	Exceeds maximum load limits
Marking	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application

4.5 Methods of inspection

4.5.1 Function test. With the nut disengaged and the handle held firmly, the blade shall fold under loads of not less than 2 pounds and not more than 7 pounds. The load shall be applied perpendicular to the blade and 1/2 inch from the tip of the blade. Noncompliance shall constitute failure of this test.

4.5.2 Strength test. The intrenching tool shall be tested in accordance with methods and procedures specified on Drawing No. 2-2-288. Noncompliance with 3.3 shall constitute a defect.

4.5.3 Hardness test. Hardness tests to determine compliance with 3.3.2.1 shall be in accordance with ASTM E 18. Three readings shall be made on the blade within a 4 inch triangle; three readings shall be made on the hinge, and three readings shall be made on the thrust washer. Any reading not within specified requirements shall constitute a defect.

4.5.4 Thrust washer and nut bond test. Thrust washer and nut sub-assemblies shall be tested by affixing the nut in a vice or fixture and applying 500 pounds of force against the inner face of the washer. Any delamination or separation of the washer from the nut shall constitute failure of the test.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or Commercial as specified (see 6.2).

5.1.1 Level A preservation. Each intrenching tool shall be completely folded and completely wrapped in a minimum 1/2 inch thick cushioning material conforming to type I or II, class B of PPP-C-843, and the wrap secured with tape, cord or twine. Alternatively, each tool shall be wrapped or sleeved in paperboard conforming to type III, style 1 or 2 of PPP-P-291. The wrap or sleeve shall be of sufficient bulk as to immobilize the tool within the box, or filler pads made from cushioning materials specified herein may be used to immobilize the wrapped tool.

5.1.2 Commercial preservation. Intrenching tools shall be preserved in accordance with ASTM D 3951.

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5.2 Packing. Packing shall be level A, B or Commercial as specified (see 6.2).

5.2.1 Level A packing. Twenty intrenching tools, preserved as specified in 5.1, shall be packed one in length, four in width, and five in depth, in a snug-fitting cleated plywood shipping container conforming to overseas type of PPP-B-601, measuring 24 inches in length, 9-1/2 inches in width and 12-1/2 inches in depth.

5.2.2 Level B packing. Twenty intrenching tools, preserved as specified in 5.1, shall be packed one in length, four in width, and five in depth in a snug-fitting tri-wall, fiberboard box, class II, style D of PPP-B-640, measuring 24 inches in length, 9-1/2 inches in width and 12-1/2 inches in depth.

5.2.3 Commercial packing. Intrenching tools preserved as specified in 5.1, shall be packed in accordance with ASTM D 3951.

5.3 Palletization. When specified (see 6.2), each intrenching tool, packed as specified in 5.2, shall be palletized on a 4-way entry pallet in accordance with load type I or Ia of MIL-STD-147, as applicable. Pallet types shall be type I (4-way entry), type IV, or type V in accordance with MIL-STD-147. Pallets shall be fabricated from wood groups I, II, III, or IV of MIL-STD-731. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means K and L or film bonding means O or P. Pallet pattern shall be number 92 in accordance with the appendix of MIL-STD-147.

5.4 Marking. In addition to any special marking required by the contract or purchase order, shipping containers and palletized unit loads shall be marked in accordance with MIL-STD-129 or ASTM D 3951, as applicable.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The intrenching tool is intended to remove earth and shall also be capable of severing roots and clearing light brush.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of this specification.
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
c. When a first article is required (see 3.1, 4.3, and 6.3).
d. Selection of applicable levels of preservation and packing (see 5.1 and 5.2).
e. When palletization is required (see 5.3).
f. Acceptance criteria required (see 6.5).

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6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

6.4 Supplementary oil treatment. A suitable oil treatment to follow the phosphate coating is No. 685K water soluble oil, supplied by McDermid Chemical Co., Waterbury, CT., or equal.

6.5 Acceptance criteria. The acceptance criteria below is recommended for use. The acceptance criteria as specified in the contract or purchase order shall be binding. Unless otherwise specified, the following acceptance criteria are in accordance with MIL-STD-105.

6.5.1 Component testing. An acceptable quality level (AQL), expressed in terms of defects per hundred units, of 2.5 is recommended.

6.5.2 End item visual examination. An AQL, expressed in terms of defects per hundred units, of 2.5 for major defects and 6.5 for total (major and minor combined) defects is recommended.

6.5.3 End item dimensional examination. An AQL, expressed in terms of defects per hundred units, of 1.5 is recommended.

6.5.4 End item testing. An AQL, expressed in terms of defects per hundred units, of 6.5 is recommended.

6.5.5 Packaging examination. An AQL, expressed in terms of defects per hundred units, of 2.5 is recommended.

6.5.6 Palletization examination. An AQL, expressed in terms of defects per hundred units, of 6.5 is recommended.

6.6 Subject term (key word) listing.

Digging
Excavation
Manual
Scoop
Shovel

6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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Custodians:
Army - GL
Navy - SH
Air Force - 99

Review activities:
Army - MD
Air Force - 84
GSA - FSS

User activity:
Navy - MC, YD

Preparing activity:
Army - GL

(Project 5120-C060)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-I-43684B	INTRENCHING TOOL, HAND, FOLDING, LIGHTWEIGHT

3a. NAME OF SUBMITTING ORGANIZATION

b. ADDRESS *(Street, City, State, ZIP Code)*

4. TYPE OF ORGANIZATION *(Mark one)*

- ☐ VENDOR
- ☐ USER
- ☐ MANUFACTURER
- ☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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